

01 • WHY THIS LAW

The public was paying for the water.

Flooding is among the costliest hazards in the United States. Damages run into the billions, and subsidized insurance made floodplain houses cheap.

That discount pulled building onto the floodplain and left the fiscal bill with the government. Risk Rating 2.0 prices the danger at the property.

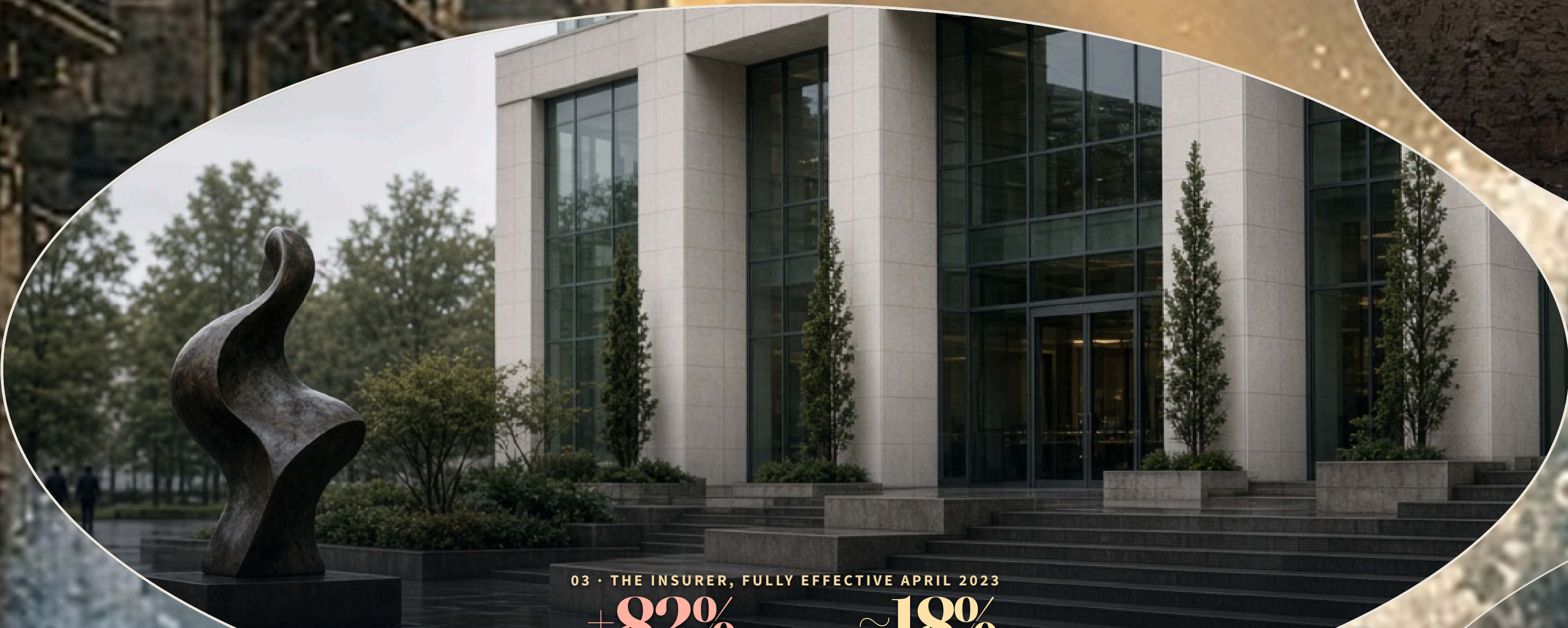
Does the shock stop at the floodplain, or does it tighten safer ground too?



02 • WHAT THE OLD PREMIUM LEFT BEHIND

A private loss, and a public one.

The price never matched the risk.



03 • THE INSURER, FULLY EFFECTIVE APRIL 2023

+82%
expected rise in the premium component
Allegheny County

~18%
FEMA's annual cap. Buyers capitalize
the whole path at once.

04 • INSIDE THE FLOODPLAIN

The price falls first. The houses leave later.

The premium lifts user cost ψ_{FP} . Willingness to pay drops. On impact the stock cannot move, so the price jumps. Construction then slows, and the price does not come back.

-28.9% **-33.5%**

price on impact
\$110,000 → \$78,340

long-run stock
4,018 → 2,686

Long-run price \$78,340 (-28.8%). Short-run stock is fixed. Safer-ground user cost ψ_{NF} does not change.

05 • SAFER GROUND

No new premium. The county still tightens.

+0.15%

Long-run price outside the floodplain, \$218,000 → \$218,326. Stock 379,126 → 379,807.

On impact the safer price is \$218,117 (+0.054%). County stock 383,162 → 382,493 (-0.17%). The new houses do not replace the ones that leave. Safer ground is already 98.95% of Allegheny.

06 • THE MODEL

One rent. Two building sites.

DEMAND - GORMAN SUPPLY - HOLMES & SIEG

$$R(S) = \frac{\alpha (W - \beta_2)}{S - \beta_1 (1 - \alpha)} \quad S = S_{NF} + S_{FP}$$

$$V_i(P) = a_i P_i^c$$

$$\dot{P}_{NF} = \psi_{NF} P_{NF} - R(S) \quad \dot{P}_{FP} = \psi_{FP}(m) P_{FP} - R(S)$$

$$\dot{S}_{NF} = a_{NF} P_{NF}^c - \delta S_{NF} \quad \dot{S}_{FP} = a_{FP} P_{FP}^c - \delta S_{FP}$$

$\psi_{FP}(m) = \psi_{FP}^0 + f m$. The shock is an 82% rise in the premium component, not $m = 0.82$.

$\psi_{NF} = 0.059 \cdot \psi_{FP}^0 = 0.1169 \cdot f = 0.0579 \cdot \epsilon = 0.08$, then $1.20 \cdot \delta = 0.02$.

Poor substitutes ($\sigma = 0.5$): the safer long-run price rises 9.9%, not 0.15%.

07 • MIAMI-DADE • AN 85% PREMIUM INCREASE

When half the county is in the water, safer ground moves.

+2.60%

X' outside the SFHA
stock +3.13%

-5.25%

A' inland flood zone
stock -6.26%

-16.68%

V' coastal high hazard
stock -19.67%

Long-run prices. About 50.5% of homes sit in A or V. Short-run prices: X +1.04%, A -6.79%, V -17.94%. Across a 15-100% rise, coastal short-run prices fall 3.7-20.5%.

08 • WHERE THE WATER ENDS

Price the risk. Make room on higher ground.

Keep property-level premiums. They pull housing off the most dangerous ground. Pair them with supply where building is safe, or the region ends with fewer houses and a tighter market.

Equilibrium counterfactuals, not causal estimates. Capitalization losses are not yet quantified. Next: every U.S. county, and an estimate of σ . Poterba (1984), Colby & Zipp (2021), Saiz (2010), Holmes & Sieg (2015).